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Infrared thermal imaging as a novel evaluation method for deep vein thrombosis in lower limbs

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Purpose: Early detection of deep vein thrombosis (DVT) is critical to prevent clinical pulmonary thromboembolism. However, most conventional methods for diagnosing DVT are functionally limited and complicated. The aim of this study was to evaluate the value of infrared-thermal-imaging (IRTI), a novel imaging detection or screening technique, in diagnosis of DVT in animal models.
Methods: DVT model of femoral veins was established in nine New Zealand rabbits. The right hind femoral vein was embolized and the contralateral one served as a nonembolized control. Measurements of IRTI, compression ultrasonography (CPUS), and angiography under ultrasonic observation (AGUO) were performed at three time points: T1 (baseline, 10 min prior to surgery), T2 (2 h after thrombin injection), and T3 (48 h postoperatively). Qualitative pseudocolor analysis and quantitative temperature analysis were performed based on mean area temperature (Tav) and mean curvilinear temperature (Tca) of the region of interest as shown in IRTI. Temperature differences (TD) in Tav (TD_{Tav}) and Tca (TD_{Tca}) between the DVT and control sides were computed. Comparative statistical analysis was carried out by paired *t*-test and repeated measure, while multiple comparisons were performed by using Greenhouse-Geisser and Bonferroni approach. Values of $P < 0.05$ and $P < 0.01$ were considered statistically significant and highly significant.
Results: Modeling of DVT was successful in all rabbits, as confirmed by CPUS and AGUO and immediately detected by IRTI. IRTI qualitative analysis of pseudocolor revealed that the bilateral temperatures were apparently asymmetrical and that there were abnormally high temperature zones on the DVT side where thrombosis formed. The results of paired *t*-test of Tav and Tca between DVT side and control sides did not reveal statistical difference at T1 (Tav: $P = 0.817$; Tca: $P = 0.983$) yet showed statistical differences at both T2 (Tav: $P = 0.023$; Tca: $P = 0.021$) and T3 (Tav: $P = 0.016$; Tca: $P = 0.028$). Results of repeated measure and multiple comparisons of TD_{Tav} and TD_{Tca} were highly different and significant differences across the T2 (TD_{Tav}: $P = 0.009$; TD_{Tav}: $P = 0.03$) and T3 (TD_{Tav}: $P = 0.015$; TD_{Tav}: $P = 0.021$).
Conclusions: IRTI temperature quantitative analysis may help further detection of DVT. Additionally, IRTI could serve as a novel detection and screening tool for DVT due to its convenience, rapid response, and high sensitivity. © *2012 American Association of Physicists in Medicine*. [http://dx.doi.org/10.1118/1.4764485]

Key words: infrared thermal imaging (IRTI), deep vein thrombosis (DVT), animal model

I. INTRODUCTION

Of all types of vein thrombosis, deep vein thrombosis (DVT) of the lower limb is the most common and fatal condition, which potentially leads to pulmonary embolism. The incidence rate of DVT was estimated to be 0.6‰ per year.[1] Since symptoms of distal DVT are typically delitescent and nonspecific, about 80% of the DVT cases, in particular symptom-free patients with distal DVT, were readily overlooked by clinicians.[2,3] A multicenter study in China showed that the 90-day incidence of venous embolism was 9.7% among inpatients with acute distress.[4] In the United States, although some 300 000 emerging cases of DVT are diagnosed annually, 3–4 times as many patients could have been underestimated owing to atypical symptoms.[5] As suggested by autopsies, DVT accounts for 72% of all deaths from a variety of diseases.[5]

Currently, angiography is the "gold standard" for diagnosing DVT but has several limitations, such as high expense, patient discomfort, invasiveness, and complications. Therefore, it is advocated that suspected patients or high-risk individuals be given a simple, noninvasive procedure to screen for or exclude DVT.[6] Although compression ultrasonography

(CPUS) remains the first-line noninvasive modality to detect DVT, it is time-consuming and demands for well-trained specialists in vascular ultrasound examination. Moreover, its sensitivity depends much on the equipments used and expertise of manipulators.[7,8]

Infrared technology has been applied in medical practices for more than 100 years, evolving from temperature-measuring thermometer in 1871 to the recently upgraded infrared thermal imaging (IRTI) diagnostic system. An early IRTI system was developed during the 1940s and applied in industry and medicine during the 1950s. In 1959, Bath first used IRTI for diagnosing arthritis and showed images of abnormally high temperature at the joints of patients with arthritis.[9] During the recent decades, along with advances in photoelectric technology, computer sciences, digital image processing, and noncooled focus plane array, infrared engineering has been developing rapidly, and thus been frequently employed in many fields of scientific research,[10–14] including medicine.

IRTI works on the principles that any object in normal state emits infrared radiant energy generated from the irregular motions of its molecules and atoms, and that more intense motion is associated with greater infrared radiation. Specifically, IRTI is a photoelectric technique in which the infrared beams from different parts of the body surface are received by infrared uncooled photography and then transferred to gray scale digital images. The resulting digital images are represented by pseudocodes corresponding to various temperature gradients, in order to indicate the range of temperature. Therefore, as a novel functional imaging detection, IRTI is able to capture surface temperature of an object by accurately measuring its infrared radiation.[15–17]

Minor functional disorders in the body's vasculature (such as a slow-down of blood flow at the very early stage of DVT) may not lead to any structural change, yet are sufficient to be remotely detected by infrared cameras, because the early pathophysiological change prior to DVT is the locally slowed venous blood flow, which in turn increases the temperature of local skin that is objectively detectable by IRTI. Thereafter, the local blood stagnation and venous occlusion occurring with a well-developed thrombolism would further alter the local temperature and intensify the thermal abnormality as detected by IRTI.

We speculated that, owing to its capacity to sensitively detect the changes in body surface temperature caused by altered blood flow in the limbs, IRTI may be used as a convenient tool to screen DVT. In this study, we aimed to evaluate the infrared presentation and distribution characteristics (PDC) of DVT in lower limbs in adult New Zealand white rabbits in which hind femoral vein thrombosis had been elicited and clearly confirmed by CPUS, so as to determine the value of IRTI in diagnosing DVT in lower limbs.

II. METHODS

II.A. Animals

Nine quarantine-certified, healthy adult New Zealand white rabbits, 4–5 months of age and 3.0 ± 5.0 kg in weight, provided by the Experimental Animal Centre of Sun Yat-sen University, were reared in single cages, fed with rabbit pellets and given free access to water for 1–2 weeks of adaptive feeding prior to the surgery. All animal experiments were approved by the Experimental Animals Committee of Experimental Animal Centre, Sun Yat-sen University, Guangzhou, China.

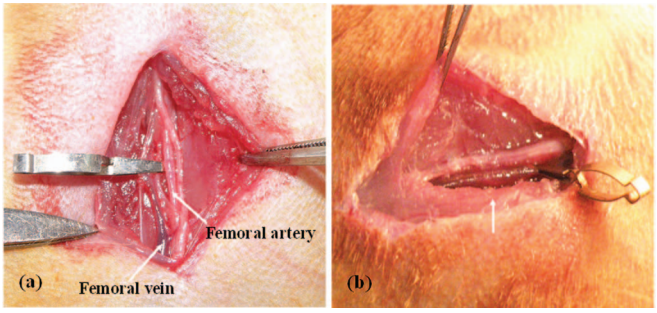


FIG. 1. The procedures for preparing rabbit model of femoral thrombosis. Panel (a) The right femoral vein was blocked by a bulldog clamp. Panel (b) Thrombosis formed at 2 h after thrombin injection, with the arrow showing that the femoral vein thrombosis was visually clear.

II.B. Modeling of unilateral femoral vein thrombosis

Enhanced methods were applied to prepare models of femoral vein thrombosis in this study.[18] First, the femoral trigone and iliac region of the rabbit were shaved for exposure of the skin, with the rabbit immobilized in supine position after general anesthesia. A 2-cm oblique incision was made outwards and downwards over the spot of strongest pulsation along the course of right femoral artery (DVT side). This was followed by blunt dissection of the external oblique muscle, the femoral artery, vein, and nerve. An atraumatic bulldog clamp was placed on the proximal part of right femoral vein to block the blood flow [Fig. 1(a)]. Next, a disposable syringe needle was used to puncture the femoral vein distally to the clamp, and then ten gentle scrapes of the inner vessel wall were applied with the needle-tip. Lyophilized thrombin powder was dissolved with 0.25 ml 0.9% normal saline before injecting 0.1 ml of the solution (with the final concentration of 200 U/0.1 ml) into the distal part of blocked femoral vein. Observation was continued, with the bulldog clamps remained in place, for 2 h until the femoral vein thrombosis were visually clear [Fig. 1(b)]. Then, all rabbits were subjected to CPUS and IRTI detections after removal of the bulldog clamps and layer-by-layer suture of the skin. The hind limbs of the rabbits were fixated by short planks and plaster bandage after DVT was confirmed by CPUS. The planks and plaster bandage were removed 48 h later, and angiography under ultrasonic observation (AGUO) was performed and IRTI detection repeated. On the left side (control side), surgical incisions were made, femoral vessels dissected and skin sutured without additional treatment. Finally, samples of bilateral femoral veins were harvested for pathological studies. The IRTI and CPUS detections were performed in 9 rabbit models at 3 time

points: T1 (baseline, 10 min prior to surgery), T2 (2 h after thrombin injection), and T3 (48 h postoperatively).

At the end of the experiments, the animals were sacrificed with a lethal dose of pentobarbital (150 mg/100 g).

II.C. Doppler compression ultrasonography and angiography under ultrasonic observation

10-MHz SEQUOIA-512 Diasonography with automatic flow-rate computation (real time convex lens probe, Siemens, Germany) was used to detect the blood flow and to confirm thrombosis in the hind limb femoral veins by 2-D color Doppler CPUS at T1 and T2, respectively [Fig. 2(a)]. The femoral vein thrombosis was detected and evaluated at T3 with AGUO. The latter shares the same manipulation as in CPUS but requires use of intravascular contrast medium, which may provide detailed information of the thrombosis further to CPUS. Since the use of contrast medium had been associated with a considerably high rate of death in rabbits in a preliminary study before the present one was conducted, we avoided AGUO at T1 and T2 until T3 when no more measurements were needed and the rabbits were to be sacrificed.

II.D. Infrared thermal imaging

The IRTI detection room was, as technically required, nondirectly ventilated and strictly temperature-controlled (ambient temperature: 20–25 °C; relative humidity: 60%). The IRTI system consisted of an ATIR-M301 sensor (AG Digital Technology Corp, Taiwan, China) with 320 × 240 pixels in resolution, and a noncooled focus plane array infrared detector which is capable of precisely measuring and storing real-time thermal images. This sensor was mounted into a thermal camera capable of snapping serial thermal images in rapid succession. The thermal camera was featured by excellent performance in temperature-sensitivity [minimum resolvable temperature difference (TD): 0.1 °C; noise-equivalent temperature difference: 0.08 °C], spectral response (ranging from 8 to 14 μm), sampling frequency (up to 30 non-interlaced images per second), and spatial resolution (capable of discriminating two dots apart with less than 1 mrad of view-angle). The thermal camera has FDA clearance for medical use.

The sensor was aligned at the same angle and operated at a distance of 80 cm away from rabbits. The surgical wound on the hind limb of rabbit was set as the focus at center of camera viewfinder, and the entire view of IRTI was zoomed just sufficiently wide to cover both hind legs of the rabbit [Fig. 2(b)]. All animal models were, respectively, infrared-captured by IRTI at three time points (T1, T2, and T3). All data collected were translated into gray scale digital images by analytical software of IRTI for qualitative pseudocolor study and quantitative temperature analysis. With different levels of pseudocode representing related IRTI images that correspond to a color scale bar of low to high temperatures (Fig. 3), the system is able to diagnose, evaluate, and analyze changes in body surface temperature according to the PDC of IRTI by different colors. We predesigned two types of region of interest (ROI):

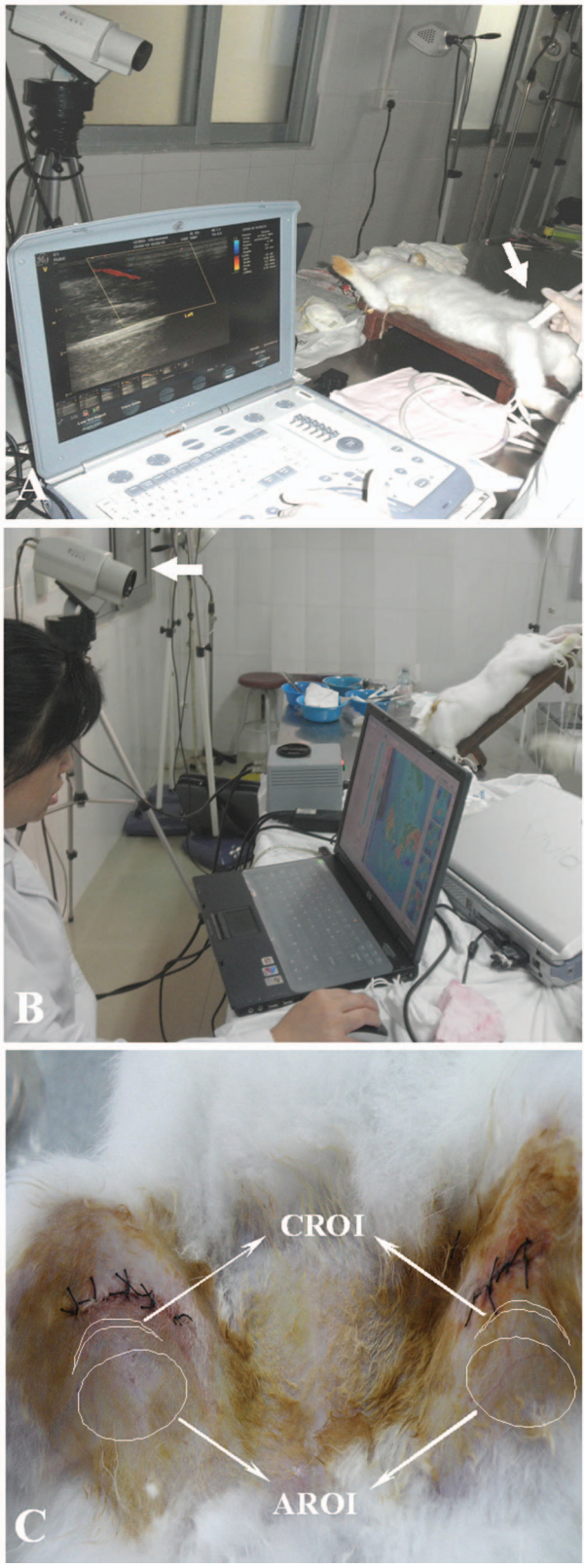


FIG. 2. CPUS and IRTI detections. Panel (a) CPUS detected the blood flow state and confirmed the thrombosis in the femoral vein of rabbit hind limb. Panel (b) IRTI: the sensor of IRTI was aligned at the same angle and operated at distance of 80 cm away from the rabbit, and IRTI images were captured and stored by the computer. Panel (c) Quantitative analysis of the extracted IRTI temperatures of ROI: Two types of ROI: (1) AROI was an ovoid, 3 × 4 cm^2 region in the inner side of rabbit's hind leg, and (2) CROI was a crescent-like region about 4 cm in arc length.



FIG. 3. Different levels of pseudocode representing related IRTI that corresponds to a color scale bar of low to high temperatures.

(1) area-ROI (AROI), corresponding to the blood-collection area of affiliated venous branches of the operation-targeted femoral vein, was an ovoid, 3×4 cm^2 region in the inner side of rabbit's hind leg, and (2) curvilinear-ROI (CROI), corresponding to the body surface projection of the studied segment of femoral vein, was a crescent-like region about 4 cm in arc length [Fig. 2(c)]. The mean temperatures of AROI (Tav) and of CROI (Tca) were extracted from captured IRTI images by analytical software system of IRTI. Temperature difference in Tav or Tca between the both sides was defined as Tav on DVT side minus Tav on control side (TD_{Tav}), or Tca on DVT side minus Tca on control side (TD_{Tca}).

II.E. Pathological study

Macroscopic pathology of the skin surface was studied before IRTI and CPUS (or AGUO) detections, respectively, at three time points (T1, T2, and T3). After T3, we harvested samples of bilateral femoral veins, which were then fixed with 4% paraformaldehyde, embedded with paraffin and stained with hematoxylin-eosin, for microscopic pathological study.

II.F. Statistical analysis

Statistical analysis was performed by using SAS (PROC MIXED in SAS 9.2, SAS Institute Inc., Cary, NC) and SPSS (PROC MIXED in SPSS 17.0 evaluation version, SPSS Inc., Chicago, IL). As stated above, the Tav and Tca at three time points (T1, T2, and T3) were extracted with IRTI analytical software. All data were expressed as mean ± standard deviation. Comparative statistical analysis of both sides (DVT and control) was performed by a paired *t*-test. The repeated-measure and multiple-comparisons of TD values (TD_{Tav} and TD_{Tca}) obtained at three time points (TI, T2, and T3) were executed by Greenhouse-Geisser and Bonferroni approaches, respectively. Values of $P < 0.05$ and $P < 0.01$ were considered significant and highly significant, respectively.

III. RESULTS

III.A. CPUS and AGUO detection

CPUS detection of vessels showed normal filling and good patency of the femoral veins on both sides preoperation. All venous lumens did not echo occluded and was readily compressible at T1 [Figs. 4(a) and 4(b)]. In contrast, the lumen of right femoral vein was rigid and not compressible with the detection probe at T2 because of blocked blood flow [Fig. 4(c)], whereas the vessel lumen compressibility and the blood flow on the control side were identical as seen at T1 [Fig. 4(d)]. During AGUO at T3, the contrast medium was visible in the femoral vein on the control side [Fig. 4(f)], but not in that on the DVT side [Fig. 4(e)].

III.B. IRTI detection

III.B.1. Presentation and distribution characteristics of IRTI in femoral vein thrombosis models

The IRTI PDC of the exposed inner sides of hind limbs were typically symmetrical on both sides at T1 [Fig. 5(a)], but definitely no so at T2 and T3. At T2, lower temperature on the DVT side was measured compared with the same side at baseline and also with the control side, whereas the temperature on the control side remained largely unchanged [Fig. 5(b)]. At T3, the both sides can be well distinguished by temperatures. There was an increase in temperature from baseline on both sides, as well as more intense PDC of IRTI (higher temperature) on the DVT side compared with the control side [Fig. 5(c)].

III.B.2. Quantitative analysis of IRTI temperatures

The Tav and Tca were extracted from the captured IRTI images by IRTI analysis software at T1, T2, and T3, respectively. Paired *t*-test of Tav and Tca between DVT and control sides did not reveal statistical difference at T1 (Tav: $P = 0.817$; Tca: $P = 0.983$), yet it showed significant differences at both T2 (Tav: $P = 0.023$; Tca: $P = 0.021$) and T3 (Tav: $P = 0.016$; Tca: $P = 0.028$). The values of repeated-measure and multiple-comparisons of TD_{Tav} and TD_{Tca} indicated significant differences and highly significant differences at T2 (TD_{Tav}: $P = 0.009$; TD_{Tav}: $P = 0.03$) and T3 (TD_{Tav}: $P = 0.015$; TD_{Tav}: $P = 0.021$) (Table I).

III.C. Pathological study

III.C.1. Macroscopic pathology on both sides

After injection of thrombin, a short dark clot was observed to develop gradually in the segment of the right femoral vein blocked by bulldog clamps. The length of the blood clot increased over time, and the vessel segment became progressively plump and dark until obvious thrombosis 2 h later [Fig. 1(b)]. At T3, the hind limbs on DVT side appeared slightly swollen in some of the rabbits yet nothing abnormal was observed on the control side in all animals.

III.C.2. Microscopic pathology of bilateral sectioned vessels

Morphology of the sectioned vessels on the DVT side under light microscope showed damaged vascular endothelial cells and homogeneously pink-stained thrombi in the vessel

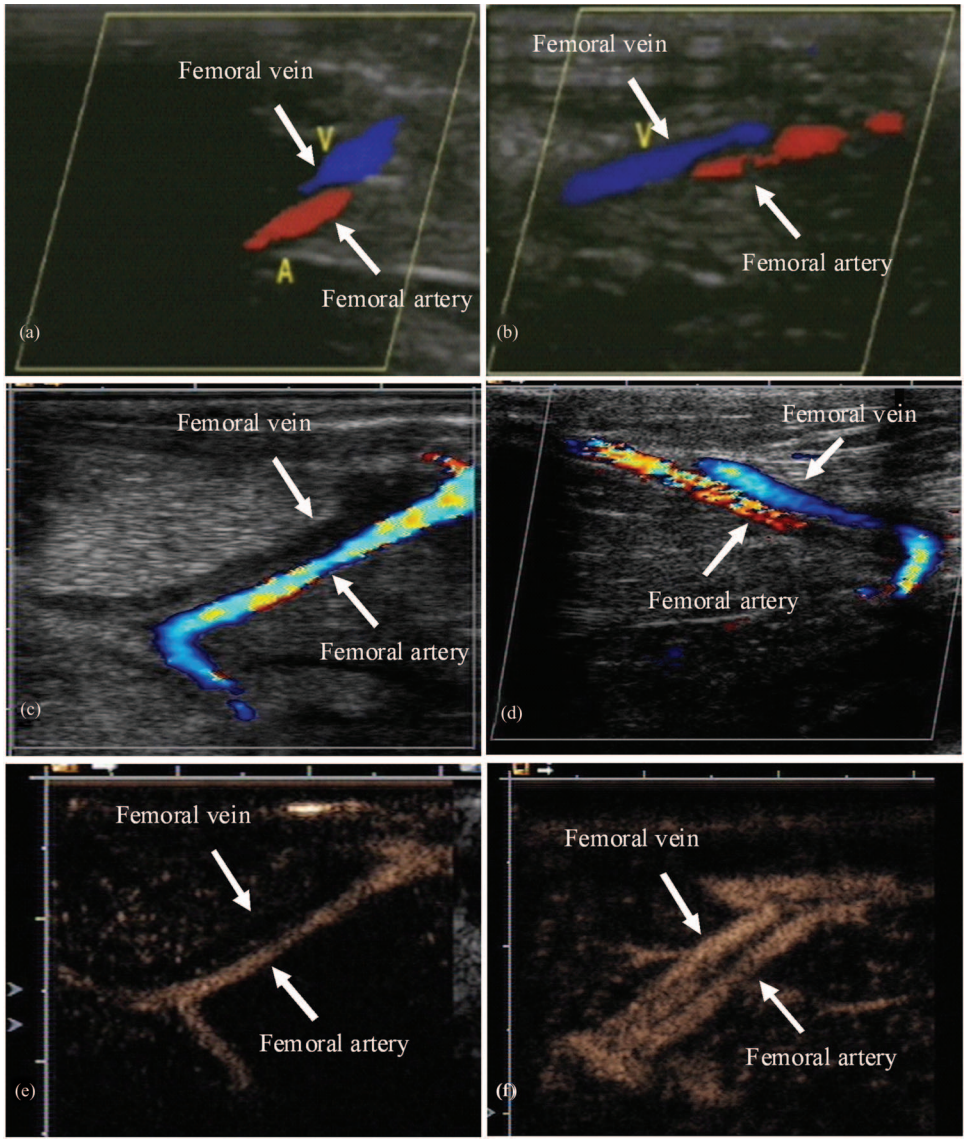


FIG. 4. CPUS and AGUO detections. Panels (a), (c), and (e) DVT side; Panels (b), (d), and (f) control side. Panels (a) and (b) T1, CPUS detection: CPUS of femoral vessels showed normal filling and good patency of the femoral veins on both sides preoperation. Panels (c) and (d) T2, CPUS detection: The lumen of right femoral vein was rigid and not compressible with the detection probe because of blocked blood flow (c), whereas the vessel lumen compressibility and the blood flow on the control side were identical with as seen at T1 (d). Panels (e) and (f) T3, AGUO detection: The contrast medium was visible in the femoral vein on the control side (f), but not in that on the DVT side (e).

lumen, while the intima of the sectioned vessels on the control side was smooth and intact with no thrombosis (Fig. 6).

IV. DISCUSSION

It is well known that the body surface temperature is closely related to blood flow volume to the skin, and that the temperature of limbs is dependent on the peripheral blood flow volume.[19] Laurence[17] based his study concerning diagnosis of thigh acute damnification syndrome on a close relationship between skin surface temperature and limb blood flow, whereby IRTI was used as noninvasive technique to detect temperature of skin surface. In his research of peripheral vascular disease through IRTI, Subramaniam[20,21] assumed that lower limb temperature was primarily dependent on peripheral vascular blood flow while limb arteries and veins played a pivotal role in heat transformation and heat distribution around the body, and in maintaining stable body temperature. Nabila,[22] who has been studying temperature regulation for years through IRTI, believed that IRTI may be used to indirectly assess the profiles of blood flow in visual and dynamic manners. However, these studies mainly addressed vessel diseases of peripheral arteries, and there has been lack of reports regarding the diagnosis of lower limb DVT with IRTI. In order to explore the potential use of IRTI in clinical diagnosis of lower limb DVT, we applied IRTI to detect lower limb DVT in rabbit models of femoral vein embolism, as compared with two other diagnostic tools, CPUS and AGUO.

Conventional theories regarding incidence of DVT were based on three aspects: (1) injury to vascular intima, (2) changes in blood flow, and (3) increase in blood coagulability. Therefore, in preparing rabbit models of unilateral femoral

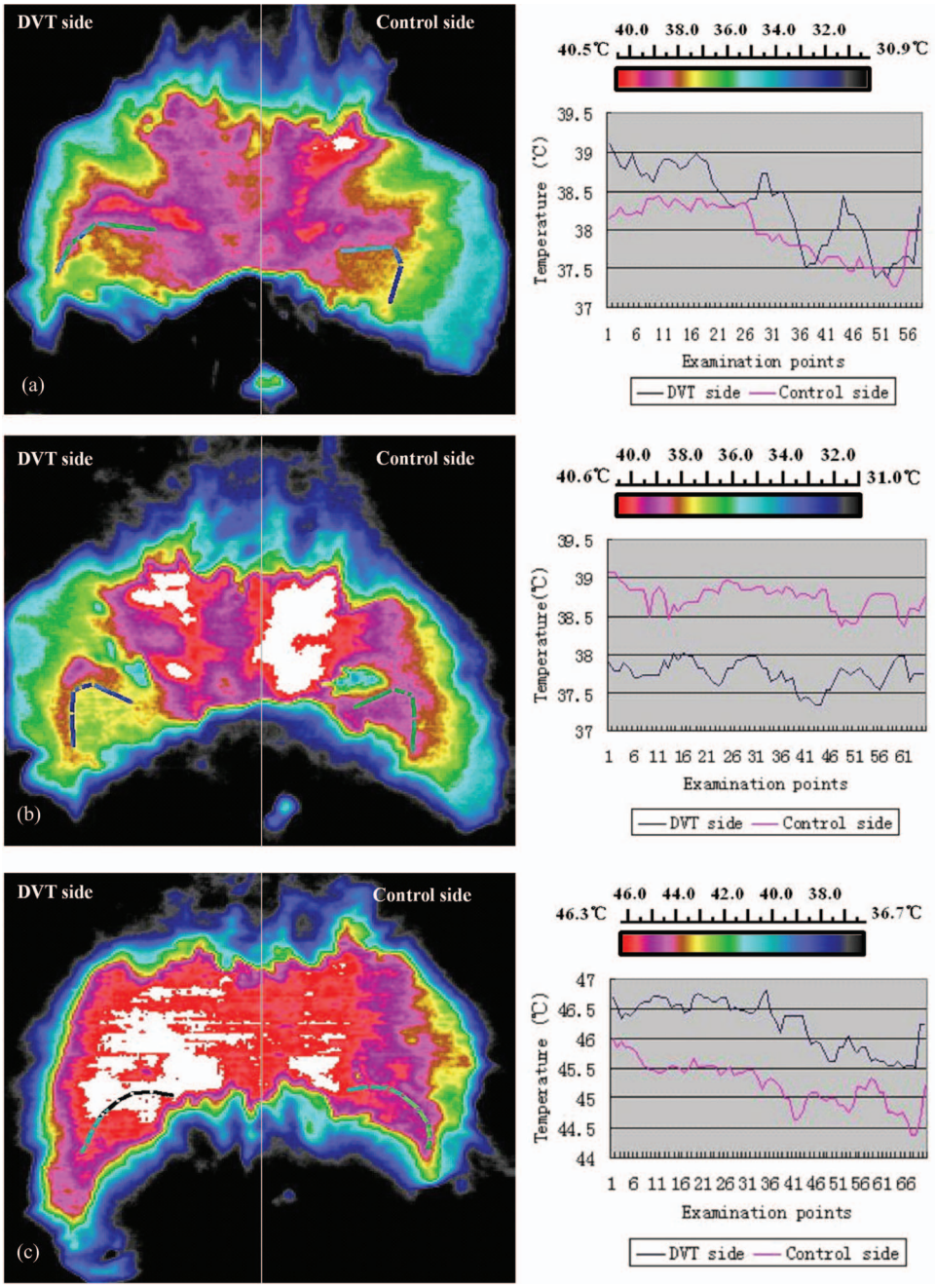


FIG. 5. Evaluation of PDC of IRTI in models of femoral vein thrombosis. Panel (a) T1: The PDC of IRTI on both sides was typically symmetrical. Panel (b) T2: The PDC of IRTI on both sides was slightly asymmetrical, with lower temperature on DVT side but higher temperature on the control side, compared with those at T1. Panel (c) T3: Bilateral temperatures were apparently asymmetrical with abnormally high temperature regions on DVT side.

vein thrombosis in the present study, we took into account these aspects to ensure successful modeling. Atraumatic bulldog clamps were used to block blood flow so as to generate blood stasis, a syringe needle employed to gently scrape the inner vessel walls to injure the vascular intima, and thrombin was injected into the blocked blood vessels for hypercoagulability. All models were detected by CPUS at T1 and T2 to confirm thrombogenesis and evaluated by AGUO at T3 to visualize the embolism.

In this study, the animal models of femoral vein thrombosis were prepared by incision, clipping, and scraping as above mentioned. All of these interventions produced local wounds. During the wound-healing process, the subsequent inflammation may give rise to increased local temperature, including the temperature stamp on the skin as measured by using IRTI. Given that inflammation starts after the surgery and lasts up to 48 h and even longer, the control side was sham-operated so that the inflammation due to healing was similar between both sides. However, within the first 48 h, by sense of pathophysiology, there should be additional factors other than inflammatory component related to wound-healing on the DVT side. These included stagnated or occluded blood

TABLE I. Data analysis comparing DVT side and control side in femoral vein thrombosis models at T1, T2, and T3 (n = 9, °C).

		DVT side	Control side	DT (DVT side−control side)	*P* value
			AROI-Tav		
T1	Mean ± Std	39.16 ± 1.75	39.14 ± 1.76	0.02 ± 0.27	$t^a = 0.239$, $P = 0.817$
T2	Mean ± Std	37.43 ± 1.65	38.05 ± 1.88	−0.62 ± 0.66	$t^a = -2.811$, $P = 0.023$
T3	Mean ± Std	41.64 ± 2.52	41.08 ± 2.46	0.55 ± 0.55	$t^a = 3.028$, $P = 0.016$
Repeated measure and multiple comparisons			$F^b = 12.424$, $P = 0.005$; multiple comparisons: $P < 0.05$ (Bonferroni)		
T2−T1	Mean ± Std	−1.74 ± 2.05	−1.09 ± 2.32	−0.64 ± 0.56	$t^a = -3.459$, $P = 0.009$
T3−T1	Mean ± Std	2.47 ± 3.29	1.94 ± 3.20	0.53 ± 0.52	$t^a = 3.080$, $P = 0.015$
			CROI-Tca		
T1	Mean ± Std	39.24 ± 1.55	39.24 ± 1.48	0.00 ± 0.25	$t^a = -0.022$, $P = 0.983$
T2	Mean ± Std	37.40 ± 1.55	37.96 ± 1.61	−0.57 ± 0.59	$t^a = -2.867$, $P = 0.021$
T3	Mean ± Std	41.91 ± 2.47	41.34 ± 2.61	0.57 ± 0.64	$t^a = 2.679$, $P = 0.028$
Repeated measure and multiple comparisons			$F^c = 11.798$, $P = 0.001$; multiple comparisons: compared T3 with T2, $P < 0.05$ (Bonferroni)		
T2−T1	Mean ± Std	−1.84 ± 2.04	−1.28 ± 1.95	−0.57 ± 0.64	$t^a = -2.638$, $P = 0.03$
T3−T1	Mean ± Std	2.67 ± 3.07	2.10 ± 3.13	0.57 ± 0.60	$t^a = 2.868$, $P = 0.021$

[a]Paired *t*-test.
[b]Repeated measure and multiple comparisons: Greenhouse-Geisser, comparison of the means of DT (DVT side minus control side); Bonferroni.
[c]Repeated measure and multiple comparisons: Sphericity; Bonferroni. AROI-Tav was the temperature average of area region of interest. CROI-Tca was referred to as the temperature average of curvilinear region of interest.

flow, tissue swelling, and local accumulation of heat following the thrombosis, which can further increase the temperature on the DVT side over time. Therefore, two measurements were performed at T2 and T3 (early and late stage of this process) in this study. According to a preliminary study conducted before the present one, thrombolysis can occur in most of the rabbits at 72 h from onset of blood clotting (data not shown). Since the purpose of the present study was to look at the feasibility of early detection (usually within 48 h from onset) of DVT with a noninvasive tool, we limited our study period to no longer than 48 h of thrombosis, in order to avert the subsequent thrombolysis that may complicate the observation.

At T3, IRTI identified higher temperatures on both sides compared with baseline (T1), but the increase in temperature was more remarkable on DVT side [from (39.14–39.24) °C (AROI-Tav: $P = 0.817$, CROI-Tca: $P = 0.983$) at T1 to (41.08–41.91) °C (AROI-Tav: $P = 0.016$, CROI-Tca: $P = 0.028$) at T3] ($P < 0.05$). In consistent with this, the IRTI-PDC on both sides was asymmetrical at T3. There was an abnormal hot spot on the DVT side where temperature difference (TD value) from the control side increased from (0–0.02) °C at T1 to (0.55–0.57) °C at T3 ($P < 0.05$). This can be explained by formation of thrombosis and blockage of venous blood return in the right femoral vein, which added to increase in body surface temperature as a result of local blood stasis. Although there was also a slight increase in temperature on control side, the temperature on DVT side increased more significantly than that on control side. Since the venous stagnation does not raise regional skin temperature immediately, 48 h postoperatively (T3) may be an optimal time point to detect DVT by IRTI and to confirm embolism by AGUO.

At T3, there was an increase in temperature from baseline on the control side, which could be arising from the local inflammation induced by surgical procedure (this may also occur on the DVT side) [Fig. 5(c)]. Importantly, the both sides can be well distinguished by temperatures, with more intense PDC of IRTI (higher temperature) on the DVT side compared with the control side. This can be explained by the complete stagnation of bloodstream, local venous occlusion, tissue edema, and probably additional inflammation following thrombosis on the DVT side at 48 h after operation.

However, at T2, lower temperature on DVT side was measured compared with the same side at baseline and also with the control side, whereas the temperature on the control side remained largely unchanged [Fig. 5(b)]. This observation may be attributed to the chronic onset of venous thrombolism. In other words, at the early stage of DVT, stagnation of the returning blood flow, local venous obstruction, and swelling of adjacent tissues may not be significant until 48 h of modeling. Besides, continuous observation for development of thrombolism with bulldog clamps remained in place may have contributed to reduced blood volume, hence lowered infrared radiation from the DVT side.

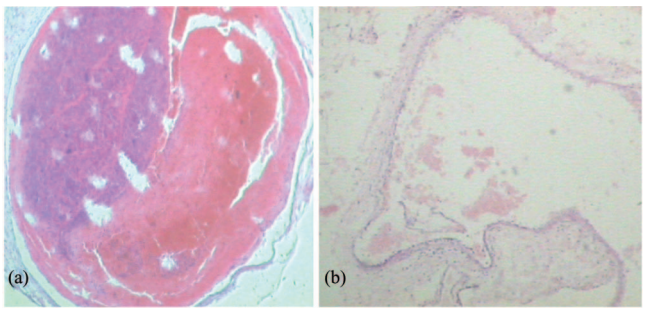


FIG. 6. Microscopic pathology of bilateral sectioned vessels. Panel (a) On DVT side, morphological observation of the sectioned vessels under light microscope showed damaged vascular endothelial cells and homogeneously pink-stained thrombi in the vessel. Panel (b) On control side, the intima of the sectioned vessels was smooth and intact with no thrombosis.

So far, reports of IRTI detection involve only peripheral arterial diseases. There has been lack for documentation regarding IRTI for detection of DVT. In the present study, quantitative temperature analysis showed that the (T3−T1) values (see Table I) were higher than those in other reports (peripheral arterial vessel diseases).[9,11,14–16] Given similar temperatures between the skin and limb vessels, assessment of the skin temperature is useful for indirect diagnosis of limb vascular diseases. It should be noted that peripheral arterial diseases typically correlate with ischemia, while peripheral venous diseases are mostly found in congestive disorders. In view of close relationship between infrared radiation and blood flow volume, it is not surprising that abnormal hypothermia was commonly reported in IRTI studies on peripheral arterial diseases while local hyperthermia was detected by IRTI in DVT, as presented in our study.

In summary, qualitative analysis of IRTI pseudocolor clearly reflected the typical infrared PDC of DVT in animal models. Furthermore, quantitative analysis of IRTI temperature may offer valuable diagnostic information for DVT. Therefore, IRTI could be used to detect abnormal arterial blood flow in lower limbs but also potential lower limb DVT thrombus.

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